

On the Method Development and Experimental Design of Ultrafast Stimulated Raman Scanning Microscopy

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Abstract

Spontaneous Raman Spectroscopy is a powerful technique for label-free imaging of biological materials. When applied to microscopy, it can produce high-resolution images without the need for an external tag. The inelastic scattering required, however, is exceedingly rare, leading to long acquisition times due to the weak signals. Stimulated Raman Microscopy addresses this by way of overlapped pump and probe beams with a frequency difference corresponding to the vibrational frequency of interest. Any inelastic energy transfer can then be measured as a change of intensity in the probe beam, allowing for high-resolution imaging via photodiode and lock-in amplifier with extremely fast acquisition times. When on the femtosecond timescale, this technique becomes a highly sensitive approach for the investigation of biological materials.

Here, we detail the ongoing development of an Ultrafast SRS Microscopy setup by way of a Nikon-Ti2 inverted microscope coupled with a galvanometric mirror apparatus. This apparatus will allow for single and multi-modal intensity acquisition rates on the scale of seconds with extremely high resolution. We detail the myriad instrumentation utilized, the optics required, and the LabVIEW programming designed to control, and more importantly synchronize, the many components of the experiment.

While efforts remain ongoing, we have achieved synchronization of our galvanometric scan mirrors with our signal detection and completed the development of our LabVIEW software suite.

Introduction

The ability to visualize molecular distributions is a fundamental requirement across the physical and biological sciences. While fluorescence microscopy remains a common practice for high-res imaging, it typically requires a process known as chemical labeling. These labels can be invasive, often inducing phototoxicity in biological samples or fundamentally altering the biochemical pathways under investigation.¹

To address these limitations, researchers have turned to vibrational spectroscopy as a label-free alternative. First experimentally observed by C.V. Raman and K.S. Krishnan in 1928, Raman scattering utilizes the inelastic scattering of photons to probe the vibrational modes of molecular bonds.² Because each chemical species possesses a unique "vibrational fingerprint," Raman spectroscopy allows for the identification of molecular structures based on their intrinsic properties, removing the need for external dyes.³

Unlike Rayleigh scattering, where the incident photon retains its energy, Raman scattering involves an inelastic collision with some energy exchange between the photon and molecule's vibrational modes. This shift in energy (or frequency) of the scattered photon can be negative, known as Stokes scattering, or positive, known as anti-Stokes scattering (Figure 1). This is represented in equation 1, where p is the pump beam and osc is the molecular vibration.³

$$\omega_{scat} = \omega_p \pm \omega_{osc} \quad \text{(Equation 1)}$$

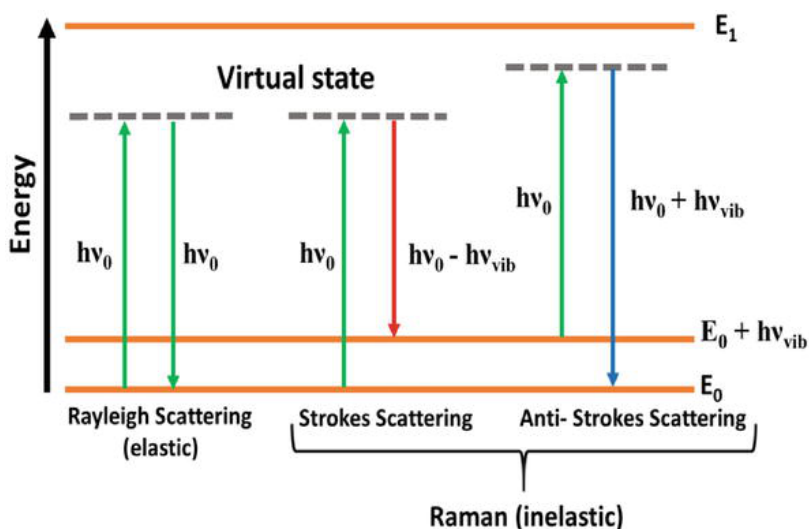


Figure 1: Jablonski Diagram of Spontaneous Raman Scattering ⁴

This inelastic scattering, however, is exceedingly rare due to the low Raman cross-section⁵, leading to long acquisition times often incompatible with delicate biological imaging. Stimulated Raman Scattering (SRS) has emerged as one way to overcome this limitation in recent years, resulting in signals enhanced by a factor of more than 10^5 .⁶

In an SRS experiment, two ultrafast beams are overlapped both spatially and temporally onto the sample. When the frequency difference $\Delta\omega = \omega_p - \omega_s$, where ω_p and ω_s correspond to pump and stokes, matches the resonant frequency of a molecular bond, the beams act as a coherent driving force, transferring energy gained or lost in an inelastic collision from the high energy pump probe to the weaker probe.⁷ Signal is then detected as a relative intensity change of either the pump beam, known as stimulated loss, or of the probe beam, known as stimulated gain.⁸

This significant increase in inelastic scattering is a result of the simultaneous interaction of the pump and Stokes photons, which creates a coherent beat frequency $\Delta\omega$. This beat

frequency acts as a driving force that resonantly excites the molecular vibrations in-phase, leading to a vastly enhanced signal compared to spontaneous Raman scattering.³

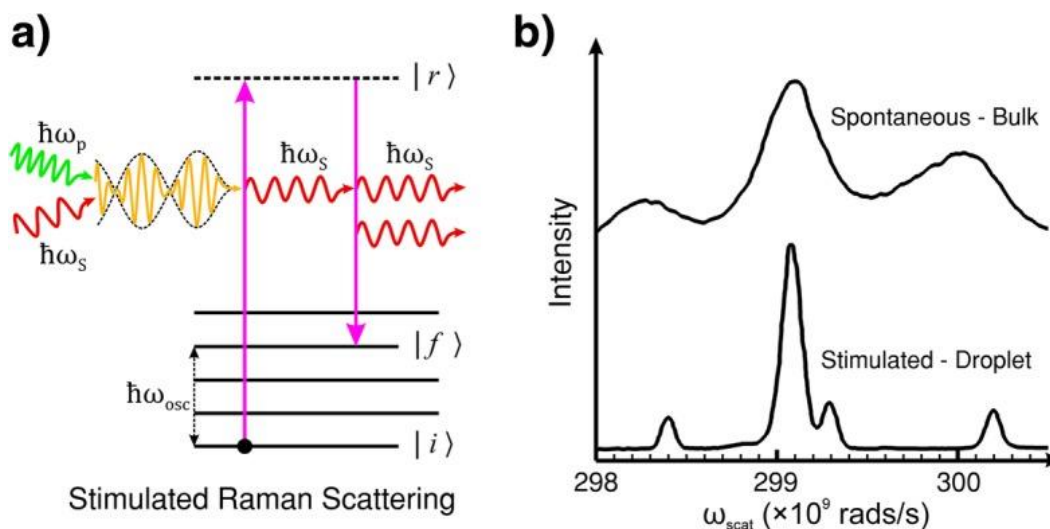
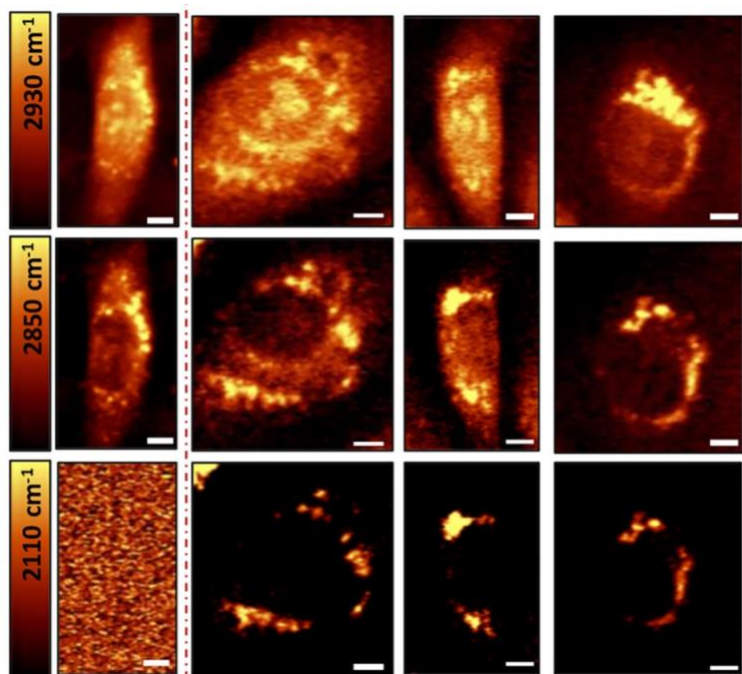


Figure 2: a) Energy level diagram of SRS illustrating the induced emission of Stokes light.

b) Comparison of spontaneous Raman scattering and SRS of bulk and droplet ethanol⁹

As a result, SRS overcomes the limitations of low scattering cross-sections and hours-long acquisition times present in spontaneous Raman. This increase in signal allows for video-rate imaging of live biological samples with microsecond dwell times. Furthermore, by utilizing a lock-in detection scheme to isolate the stimulated gain/loss, SRS effectively eliminates the autofluorescence background that often overwhelms spontaneous signals in complex tissues. Because of this, SRS has been utilized in a large range of experiments requiring biological imaging, including the study of lipid metabolism. SRS allows for the detection of neutral lipid distribution, identification of lipid droplets and subcellular levels and accurate quantification of lipid levels *in vivo*. This is through the presence of carbon-carbon double bonds in unsaturated lipids, enabling for the differentiation of unsaturated lipids in living cells.¹⁰ Figure 3 shows an

example image output of human aortic endothelial cells via the C-D vibration in the $d_{34}OA$ molecule.¹¹



*Figure 3: SRS images of HAEC cells*¹¹

The technique is also heavily utilized in inorganic and nanoparticle fields, with the potential to be applied for spatial mapping of microplastics in tissue. These plastics on the micrometer and nanometer scale are extremely prevalent in the modern world, with quantification suggesting more than 10^5 particles in each liter of bottled water.⁹ With its ability to select single-modal stretches with no autofluorescence background, SRS has been utilized to study such particles in a myriad of samples.¹² Upon completion of the experimental setup of this experiment, it will be utilized to undergo one such study on human lung samples.

Materials and Methods

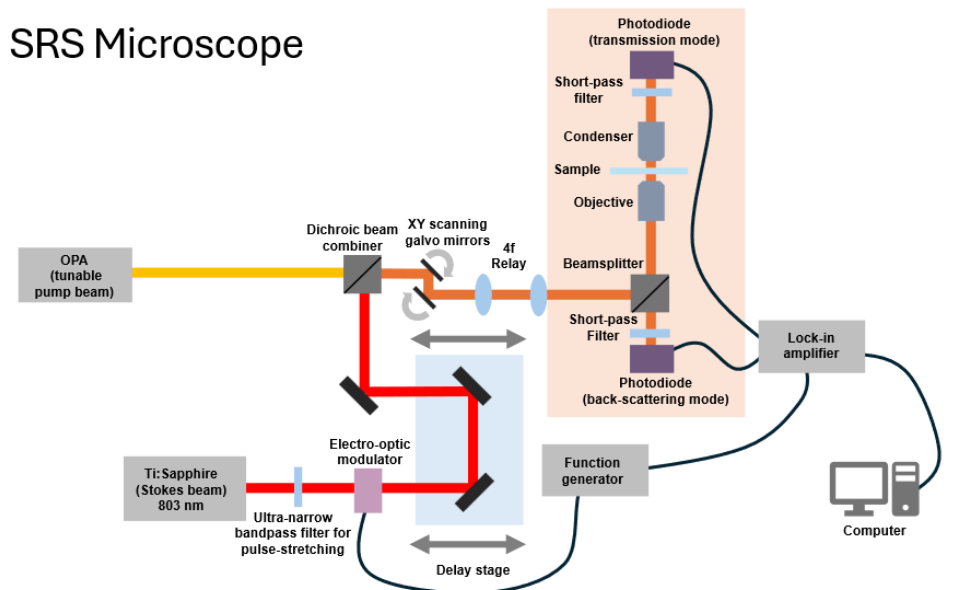


Figure 4: Experimental Setup Scheme

Optics

An ideal excitation scheme for an SRS experiment, especially in the context of scanning microscopy, requires spectrally narrow pulses that are overlapped both spatially and temporally.⁴ In the context of ultrafast SRS, pulses deliver extremely high peak power but with extremely broadband signals as a result of the time-frequency uncertainty seen in Equation 2.

$$\Delta\nu\Delta t \geq \frac{1}{4\pi} \quad (\text{Equation 2})$$

This can lead to the undesired excitation of multiple vibrational modes, obscuring the signal desired in single and multi-modal schemes. As a result, significant spectral focusing is required for ultrafast pulses.⁶

With a diffraction limit of 300nm, as determined by Equation 3 (where NA is the numerical aperture of the objective), the spatial overlap of our beams is of high import. Should the pump and probe beams fail to overlap, we risk misattribution of our intensity readouts, improper alignment of the galvanometric mirrors, and inaccurately enhanced / diminished intensity shift readouts.⁶

$$d = \lambda/2NA \quad \text{(Equation 3)}$$

A Spectra-Physics Solstice Ace Titanium-Sapphire Laser first generates a beam at 803 nm. Part of the beam is then tuned via a TOPAS optical parametric amplifier (OPA), serving as our pump and the diverted 803 nm beam is our probe. The probe beam originates with a beam width of ~12mm whilst the TOPAS pump beam has a beam width of ~4mm, as measured via a razor edge spot test. To account for this size discrepancy, a Galilean telescope constructed with a 300mm convex and 150mm concave telescope, separated by 150mm, is utilized, reducing the probe beam size by a factor of 1/3. This smaller width also assists scanning ability, as the scan angle range of the galvanometric mirrors increases with decreasing beam size.

To account for its large bandwidth, the probe beam then encounters an ultra-narrow 803nm bandpass filter, which transmits only light of $\lambda = 803nm$, with a spectral selectivity of 0.5nm. This temporally stretches our pulse and thus greatly narrows its pulse width (Equation 2).

The probe beam then encounters a 90/10 beamsplitter, splitting off a small portion of the beam for a single shot autocorrelator from Miniopic. The unimpeded beam then encounters an Electro-Optic Modulator, which will be discussed in the future works. A time-

delay, consisting of a manual X-Y stage equipped with two reflective mirrors, is utilized to ensure temporal overlap of pulses. This is done by altering their path length by linearly moving said time-delay, increasing the path length of our probe beam by $2\Delta X$. Should significant dispersion be introduced by way of further refractive optics, this stage would help to correct such effects.

The beams are spatially overlapped via a dichroic mirror, with a 0 angle of incidence for probe and a 90 angle for pump. The overlapped beams are aligned onto the center of the Y axis mirror of the galvanometric mirror. It is then reflected upwards a distance of 10.5mm onto the X axis mirror.

The rotating nature of the galvanometric mirrors means that, though initially aligned to the center of the back aperture of the objective, beams quickly walk off as mirrors rotate. Figure 5 shows the 4f lens relay utilized to address this, a practice commonly utilized to account for galvo tilt in SRS systems.

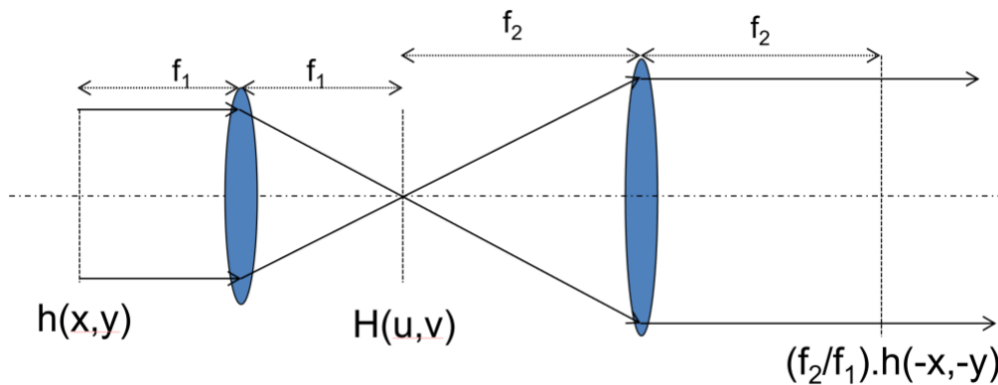


Figure 5: Fourier Relay¹³

The relay consists of a pair of chromatic doublets with focal lengths 50mm and 200mm, separated by $(f_1 + f_2)$. The galvos are placed (pivot point) f_1 behind this relay and the image plane f_2 after. Upon encountering the first mirror, the beam undergoes a spatial Fourier transform, converting the change in angle into a change in position. As a result, all incident beams are parallel to the optical axis but shifted spatially. The second lens, oriented oppositely, reconverts this change of position into a change in angle¹⁴. In addition, these lenses serve to magnify beam size by a factor of 4x as determined by the equation $M = \frac{f_2}{f_1}$. This ensures that both beams fill the back aperture of the Ti-2 objective, thus ensuring proper focusing onto the sample. Beams enter the back aperture of the Ti-2 and travel approximately 200mm before being reflected upward into the objective. Depending on the desire for spontaneous or stimulated Raman, beams encounter a long pass or short pass filter. For spontaneous Raman, this ensures detection of solely the Raman-loss pump beam.

Galvanometric Mirror Setup

While single-point microscopy can be a powerful technique, a broader picture is needed to effectively image myriad samples. For this, we transition to scanning microscopy, where a series of points of the sample are imaged. This can be done by a) moving the sample along the X-Y plane, usually via a piezoelectronic stage, or b) moving the beam via rotational optics.

Whilst the use of a piezoelectric stage can provide extremely accurate spatial imaging, galvanometric scanning is beneficial in its extremely low comparative cost and increased speed. In addition, with a larger beam size, the minute accuracy of the piezoelectric stage is not required.

As such, option b was adopted, with a fixed sample and rotating beam. To do so, the Thorlabs GVSK2-US2D Galvo System was utilized (Figure 6). The GVSK2 consists of two single axis systems of motor and mirror assembly, mounting bracket, driver and heat sink. Each is equipped with silver-coated mirrors. The term “galvanometer” refers to the precision motor whose acceleration is directly proportional (with adjustable scaling) to the current applied to the motor coils. When current is applied, the motor shaft rotates through a limited arc.

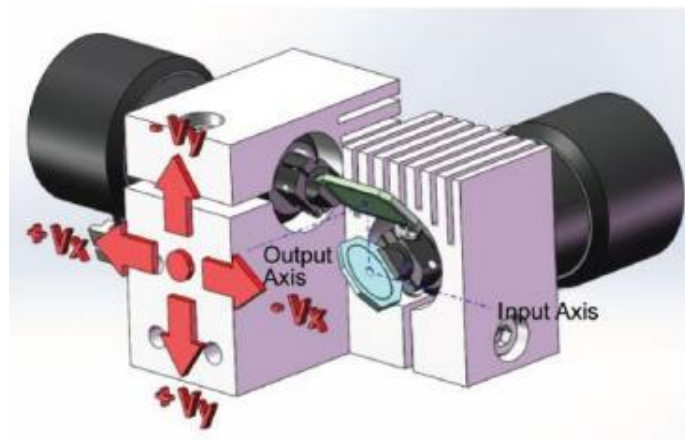


Figure 6: Model of the galvanometric mirrors and possible motion

The Thorlabs GVSK2 motors possess a couple of important specifications worth discussing here. The first is their two driver boards, J9 and J10, for connecting to the galvanometric mirrors and some voltage generator, in this case an NI-DAQ Data Acquisition card. Another is the jumper, JP7, capable of tuning the volts/degree scaling factor. This allows for refinement of scanning ranges depending on the area of sample to be imaged. For smaller ranges of sampling, a higher V/D scaling is preferred to ensure precise motor movement. For initial testing, the smallest V/D of 0.5 was utilized as it allows for the largest possible scanning range.

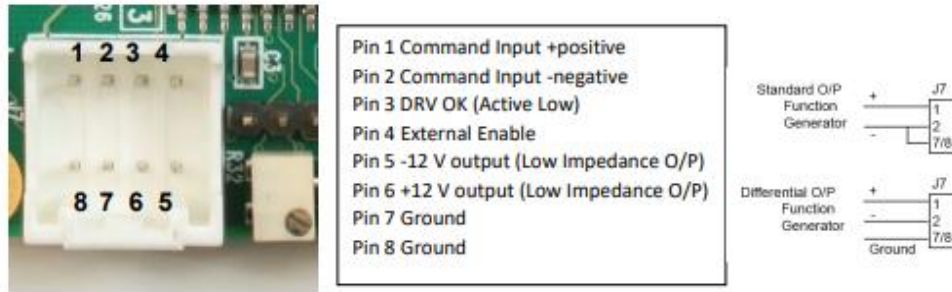


Figure 7: J7 Command Input Connector Pin identification

Both motors are powered by a GPS011 power supply. Both motors and DAQ are grounded by way of our laser table tested for ground.

Data Acquisition Card

SRS setups contain several moving parts that need to be timed exactly and often in reference to one another. To achieve this, the USB-6421 OEM Data Acquisition Card (DAQ) from National Instruments was utilized. The USB-6421 is a USB Type-C bus-powered mioDAQ device. It features analog input channels with four input range settings from ± 200 mV to ± 10 V, and analog output channels that can output ± 10 V with update rates of 250 kS/s/ch. Additionally, the USB-6421 digital I/O channels can function as static input or output channels, hardware-timed channels up to 10 MHz, or as triggers, clocks, or I/O for the four onboard.

This DAQ contains many features key to the synchronization of the experiment. These include a series of Analog Output channels to provide rapid and accurate voltages to the galvanometric mirrors, a Digital Output channel configured as a Counter for the spectrometer/PIXIS external trigger, and a USB-C connection to a computer, which also provides

its power. The DAQ is housed in a self-designed casing that protects the vulnerable OEM card from dust and outside interference whilst providing openings for various wirings (Appendix).

Before detailing specific channels and wirings, it is important to understand what exactly a DAQ does. The card is equipped with a series of “channels”, which correspond to some of the physical pinouts on the card (see Channel Outputs). Each channel, in essence, operates an individual motor/machine. For the analog output channels, the card utilizes a digital to analog converter to convert between digital inputs (namely, some voltage specified on a program) and a continuous voltage output. This allows for precise, software driven (through the DAQ hardware) control of any voltage outputs. The card can also produce digital outputs, skipping any conversion to voltage, and just producing an On/Off signal for some hardware. Often referred to as a Counter, this pulse train (which is, really, a continuous series of alternating on and off signals) can be used to control cameras, spectrometers, or other hardware that requires precise timing as to when to fire/activate. The card is also equipped with input channels, which can be used to gauge the status / position of hardware, though this is not utilized in the current experiment.

The DAQ also contains an internal clock of impressive speed, with a maximum sample rate of 250 kS/s. This can be used to synchronize various outputs, creating precisely sequenced outputs through various channels to run an entire experiment in tandem.

Channel Outputs

The DAQ contains over 50 pinouts with various assignments ranging from analog inputs, outputs, ground, and chassis ground. We utilize the Digital and Analog output pinouts, as well

as Analog Input. Importantly, an additional 50 Position Receptacle Connector from Digikey was placed atop the Analog/Digital output pinouts. This ensured protection of the board circuitry, minimized any possible static noise that may arise, and ensured that pinouts could be easily changed if needed by resoldering. As the X and Y mirrors of our galvanometric mirror systems operate independently, each needs its own channels and the voltage outputs they produce. Each galvo mirror utilizes two input wires and is equipped for voltages ranging from -10V to 10V. For each, a wire is soldered to an analog output pin and its respective ground to minimize noise in the signal. The X and Y galvos are wired to pins (3,4) and (5,6) respectively.

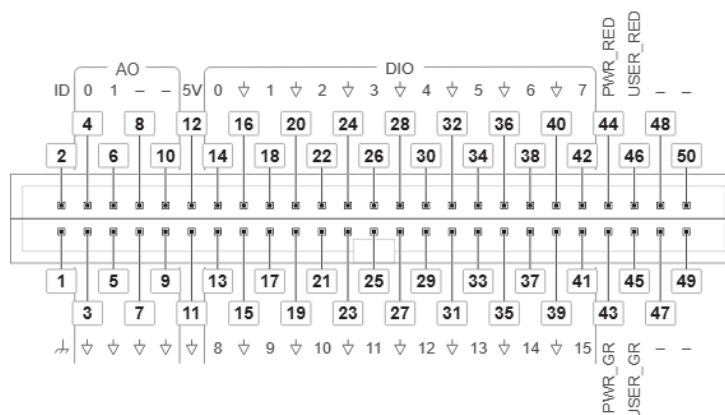


Figure 8: DAQ-USB-6421 AO/DO Pinouts

For our PIXIS, the digital output channel and ground corresponding to pins (14,16) are soldered in the same way. This connects to the PIXIS External Trigger. This wire, specific to the PIXIS camera, has a BNC cable output, and thus required the use of a BNC - double wire adaptor. Whereas the galvo's analog output corresponds to voltages, this channel provides a simple trigger in the form of a brief logic on signal to inform the camera to begin acquiring. In similar setups such digital output is commonly referred to as a counter pulse.

LabVIEW

Released in 1986 by James Truchard and Jeff Kodosky as the flagship product of NI Instruments (who also made the DAQ), in an effort to minimize the time needed to program instrumentation systems¹⁵, LabVIEW has become an extremely popular option amongst engineers and scientists. Sporting a unique graphical interface, its unique wire-like data handling made it perfect for programming and interfacing with the myriad hardware used in a laboratory.

The basic structure of Labview is as follows: A simplified “front end” containing buttons, switches and displays, and a “back end” containing the myriad logic gates / for loops / data flow that enable the front-end operation. Along with simplifying data acquisition in processing through a clear separation of input and output, this allows those less familiar with computer programming the use of complex programs without the usual barrier of entry.

There were three main areas of focus for this programming project. 1) integration and automation of important Nikon Ti2 Microscope controls, 2) precise control and automation of the galvanometric mirrors, and 3) the automation and synchronization of data collection. Step 3 is acquisition by way of a photodiode + function generator. This section will detail the work done thus far as well as the proposed steps to come.

LabVIEW Basics

Before jumping into the more complicated aspects of the program, it is important to outline the general basics of programming in the rather esoteric language. By default, LabVIEW

progresses from left to right by way of the yellow error wire. Along with detecting and conveying errors, this wire tells both LabVIEW and the user the direct order to operate in. Similarly, all data in LabVIEW travels by way of wire. The input of, say, a multiplication function, or “block”, is two wires, and the output wire is the product (Figure 9).



Figure 9: Example LabVIEW Logic

Also important to understand is the use of sub-VIs, which are almost analogous to python functions, but more self-contained. These are sub-programs that another program can call upon to complete certain tasks, and can be fed inputs of any kind and similarly produce a myriad of outputs. These serve to reduce the computational load of one program, whilst also helping with debugging. Though it will be revisited later, an example can be seen in Figure (9), where the block titled “Reset” is a sub-VI, and the various wires leading into said block are its inputs. The logic flow can be followed using the yellow error wire, which moves from the left of the figure to the right, exiting the sub-VI once complete and reporting any errors that may have occurred.

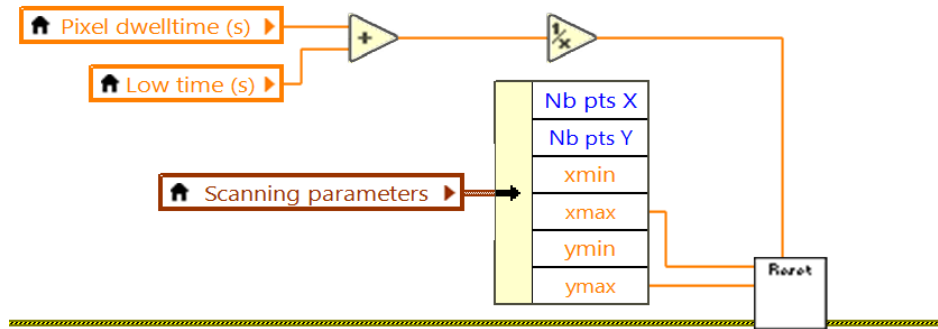


Figure 10: Sub-VI “Reset” Inputs and Outputs

Front-End

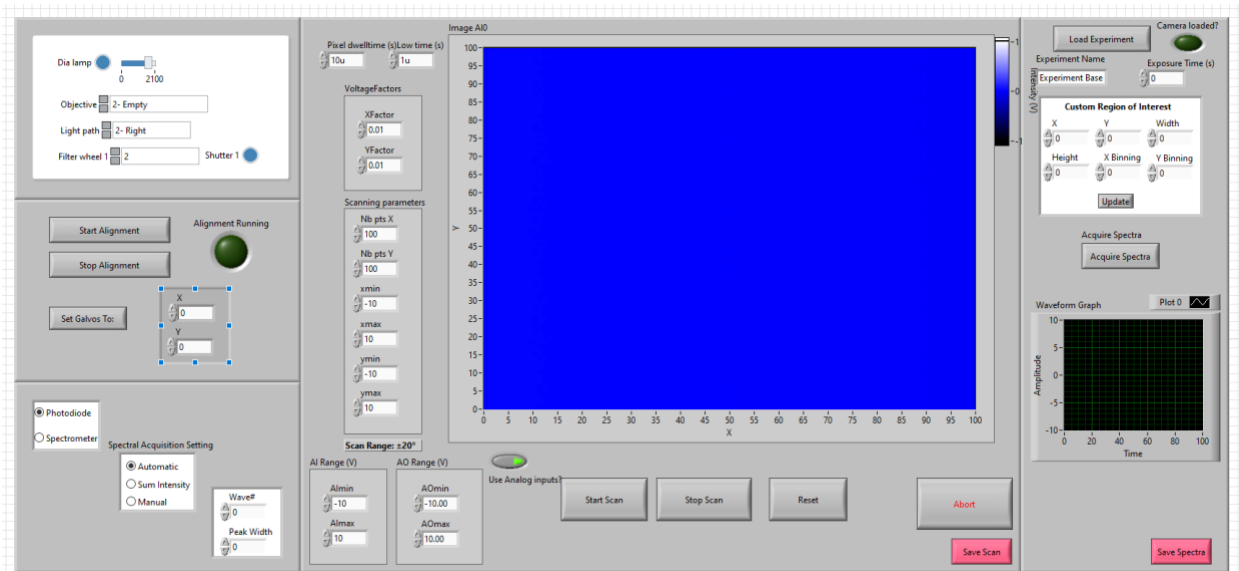


Figure 111: Front End GUI of LabVIEW Program

Figure 11 displays the front panel of the LabVIEW code, which is what is accessed when not editing or debugging the back-end code. This is what anyone utilizing the experiment would

interact with, avoiding the complicated appearance of the back-end. It is, mostly, divided into sections based on the instrument buttons/settings control. In the top left is microscope settings, where LabVIEW displays the current setting of the objective, light path, filter wheel, shutter, and lamp. The lamp can be turned on and off, as well as having its intensity altered.

Directly below the microscope section is the alignment/direct galvo control. When aligning the galvanometric mirrors to the back objective and/or the Fourier objectives, it is important that alignment occurs at the position corresponding to the (0,0) of the mirrors, else a risk of immediately misaligning when supplying voltages. As the mirrors are rather loose in their mounts to allow for free range of motion whilst active, this helps to ensure the above statement while adjusted by hand. This section also includes a button to set the galvo mirrors to specific, static positions. This will prove integral when obtaining distance calibration, which will be detailed in future works.

Below that, still along the left-hand side, is the section containing the data acquisition settings. For acquisition by way of photodiode, the other two clusters (Multi-Modal Acquisition Settings and Manual Settings) are superfluous. When set to spectrometer, users must specify their desired data acquisition settings. If this setting is manual, the desired pixels (which correspond non-linearly to wavenumber) and peak width must be specified. This is, in effect, the user specifying which peaks (modes) to consider when imaging. All of these settings should be set before beginning the scan.

Moving to the right, the large center block handles the scan and displays the intensity graph output. Here, the user can specify a number of settings for the scan acquisition, most

importantly the pixel dwelltime, X min and max, Y min and max, # of X points and # of Y points. The user can also specify if they wish to “Use Analog Inputs”, which for the single-modal scheme, essentially translates to recording signals. If not selected, galvanometric motion will occur without acquiring signal input. Here the user can also start, stop, or reset the scan/galvos. Finally, the user can save their scan once completed via an inputted file path.

The rightmost block contains controls relating to the spectrometer/PIXIS camera. The user can specify and load a new or past experiment, as specified by the experiment title. This allows the user to resume past experiments quickly. Upon loading an experiment, the set exposure time is loaded and displayed, but can also be changed at any time. If an experiment is not loaded before a scan is begun, this step is triggered automatically. The user can also specify a region of interest on the PIXIS. They can, finally, acquire and display sample spectra from the spectrometer to analyze peaks and spectral profiles.

Back-End Organization

Acting as a central hub for the numerous instruments utilized in the experiment, the overarching LabVIEW project can be difficult to follow. As such, it may be helpful to detail the overarching back-end structure beforehand.

The program dwells in a two-frame sequence: initialization, and running. Somewhat contrary to its title, the initialization does not address the galvo or spectrometers’ initialization directly: instead, it queries the microscope and reads out its current settings to ensure an accurate front-end readout. It also, importantly, opens a queue, which will be utilized to communicate between front-end Booleans and resulting actions, or more scientifically,

producer and consumer loops. During the running of the program, two while loops run simultaneously. These are the producer and consumer loops, where one loop deals with user-inputted changes, usually the changing of a setting or pressing of a button. These changes produce an event, which, in the case of a setting change, is rather simply handled. For more complex operations, this loop sends a command to the consumer loop to execute some sub-VI or complex function. This communication occurs by way of a queue, which was created in the initialization sequence, that transmits commands across loops. If the producer loop does not detect a user-inputted change, it idles, updating any microscope values. If the consumer loop does not receive a command, it idles. The ABORT button stops all loops and ends the program.

Microscope Integration

This section was largely adapted from code for the Ti2 microscope developed by Dr. Sebastian Mailfert, who provided a copy compatible with the 2025 LabVIEW version. It first utilizes an Open Automation block which, when supplied with the tag communicated via the cyan wire, obtains a reference to the Ti-2 microscope. This tells any resulting functions that utilize the Ti-2 exactly how to access it. The Nikon_CheckStatus sub-vi then utilizes that reference to obtain the current status of the microscope and its myriad components. Depending on the add-ons utilized, this can be altered to include any additions to the experiment. Sub-vi Cluster_MAJ then packages those components into a cluster, which can be understood as a package of different variables and their values, that can be called upon when changing any settings on the Ti-2 microscope.

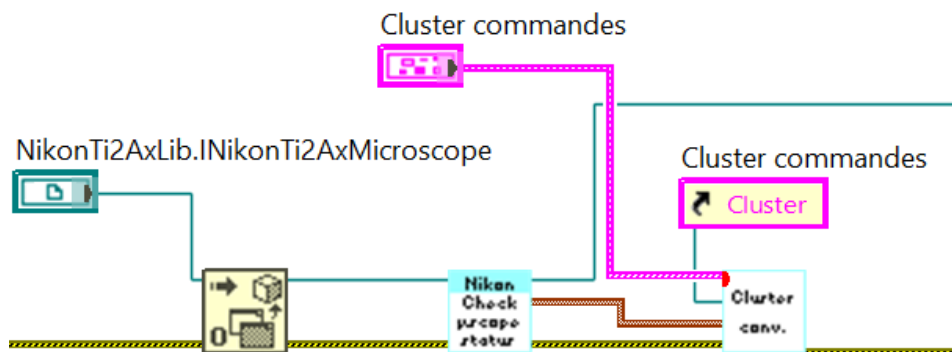


Figure 12: Microscope Initialization

When it comes to altering microscope values, the Ti-2 utilized for this experiment lacks many add-ons that can be controlled via LabVIEW. As a result, this code is most useful in displaying the current settings of the microscope, that being Objective, Light-Path, Filter Wheel and Shutter. We can control the Dia Lamp and adjust its intensities from the front-end diagram, which is handled in two event cases of the producer loop (Appendix).

Spectrometer Integration

Outside of its use in multi-modal acquisition, the spectrometer/PIXIS can be controlled by way of the Lightfield LabVIEW package, where Lightfield is the original software written for the PIXIS. Equipped with a number of sub-vi's, one can control several features from our central LabVIEW hub. For those features outside this package, it is relatively simple to construct our own.

By calling up Cam Init.vi, the user can load any “experiment” as specified by name. Note that it is of great import to release any instruments allotted for one experiment before loading another in Lightfield. This will load the desired “experiment”, which can essentially be

understood as a cluster of acquisition settings. This loading will acquire and display the acquisition time. Through another event case, the user can also change this setting. Note that acquisition time is the period in which the spectrometer acquires per image.

Though rarely employed, the user can also specify a region of interest through event case 11 (Appendix). This allows for a specific, user-defined range of wavelengths (or frequencies) within the full spectrum to isolate for analysis.

Lastly, the user can acquire and display a single diagnostic spectrum in order to determine the wavenumber of the bond stretch of interest through event case 13 (Appendix). This remains displayed as scan acquisition is ongoing, providing a helpful reminder as to the bond (or bonds) being imaged. This spectrum will auto-adapt to the axis settings specified in Lightfield.

Ni-DAQ Integration

The NI-DAQ USB 6421 OEM comes equipped with a very thorough LabVIEW package. In order to clearly communicate the more complex single and multi-modal acquisitions, it is important to first detail a number of integral NI-DAQ sub-vis. Note that the inner workings of these sub-vi's are not needed to understand their function and as such will not be detailed here. An image of each sub-vi can be found in figure 13.



Figure 13: Ni-DAQ Sub-vi's

First is the Create Channel sub-vi. Detailed in the DAQ section, a channel can be understood as corresponding to a single pin-out on the DAQ, with the exception of internal clocks. These channels can be configured in a number of ways, including Analog Input, Analog Output, Digital Input, Digital Output, Counter Input and Counter Output, and the assignment of channels is often the first step in scanning programs.. Analog channels require a specified minimum and maximum voltage, which can be tuned to fit experimental needs / increase precision. This sub-vi outputs a task, which looks like a squiggly purple wire, that contains all information about a channel.

Commonly paired with the Create Channel vi is the Sample Clock.vi, which specifies the scanning rate, number of samples, type of sampling (finite vs continuous). When fed a task, it imparts this timing information onto it, outputting the updated task. Scans that utilize this clock are referred to as hardware-timed, as the DAQ produces the synchronization scheme.

These tasks are not, by default, automatically triggered when called upon. Instead, we must start each task with the Start Task.vi, which gets any hardware ready for initiation. Buffers are created and counters or clocks are prepped for their trigger. Once armed, Read and Write can be used to send or acquire voltages. After a task has begun and its usefulness has ended, we can use Wait Until Finished, Stop Task, and Clear Task to avoid any mess.

Scanning

With an understanding of these sub-vis, scanning becomes rather simple. Adapted for our experiment from a more general scanning program¹¹, the program first creates and specifies channels, which consists of AI input (lock-in amplifier), AO output(galvos) and a pixel clock (position -> signal synchronization) (Figure 14). These channels operate at a rate of $1/(\text{pixel dwelltime} + \text{low time})$. It also specifies voltage ranges, which in practice will rarely change, but can be used to tune the sensitivity of the voltage in/outputs.

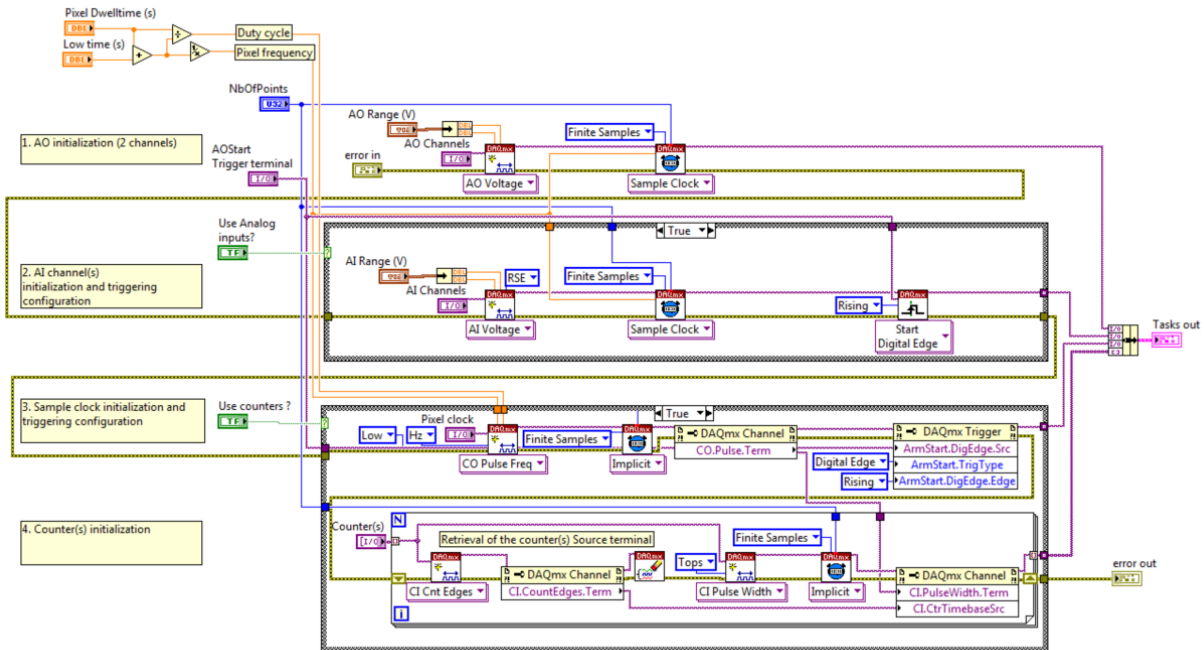


Figure 14: Init.vi ¹⁶

With channels + tasks created, it transitions to a scan, which proceeds via the various DAQ sub-vi's. Note that the three case structures reliant on "Use counters?" are to be utilized for pulse synchronization involving a function generator and electro-optic modulator, discussed more in section 5. The program first arms the AI and AO tasks then generates a raster scan pattern. It uses DAQ_Write.vi to start the AO voltage generation and DAQ_Read to measure the intensity readout of the photodiode + lock-in amplifier at each point. Once complete, it outputs the data (in the form of a 1D array) to be handled in the main vi and end any tasks.

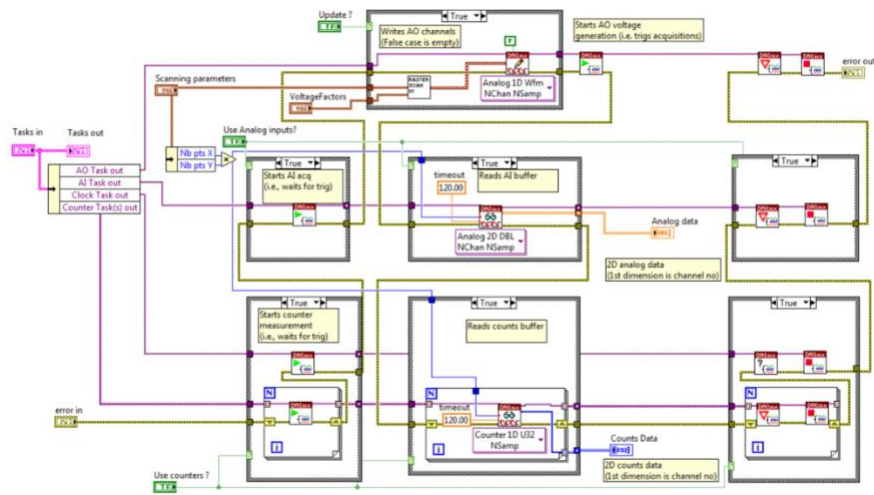


Figure 15: Scan.vi¹⁶

As the data output is 1D, some reshaping is needed in order to display an intensity graph that corresponds to the sample, which can be seen in Figure 16. This can be done by separating the output array into its constituent inputs (if more than one input was utilized) and reshaping it into the form of # of points X by # of points Y.

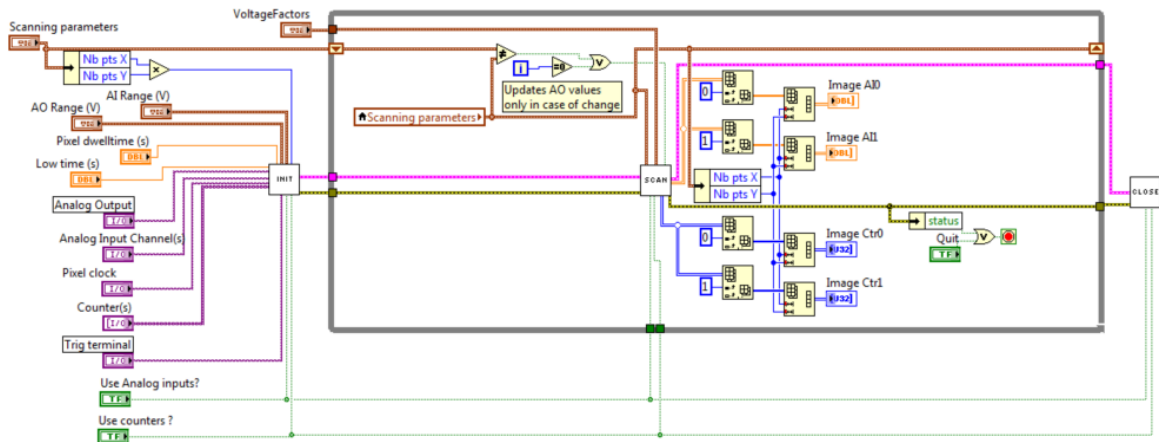


Figure 16: Block diagram of the integrated program GPScan.vi¹⁶

Galvanometric Control

With an understanding of the various NI-DAQ sub-vis, understanding the galvanometric programming becomes rather simple. Program Alignment.vi is triggered by the “Start Alignment” button, which queues a command to the consumer loop to begin the sub-vi. While active, the green boolean (light) lights up. It can be stopped, somewhat obviously, by the Stop Alignment button, which sets the Stop FGV to true.

It operates by creating a channel, which corresponds to the physical pinouts the motor are soldered to. After informing the main program that alignment is running by way of the Align. Running FGV, it enters a while loop that constantly feeds the galvo’s voltages of 0, which correspond to the (0,0) position. Should the stop alignment button be pressed on the main program, this loop stops by way of the Stop FGV, stopping and clearing its task before exiting.

The simple galvanometric control can be understood in a very similar way, just lacking the While Loop. It is also handled completely in the producer loop. It creates the channels, unbundle the desired positions from their cluster, and write those positions to the galvos. Afterwards, it stops and clear any tasks, letting the user know that the desired position has been achieved.

The reset function operates in a similar fashion with some added safety measures as a result of the scan often ending at values deviating heavily from (0,0). To avoid a large “snap-back” of the galvos, it ramps downward from our final positions (X, Y Max) to (0,0) at the scanning rate.

Spectrometer Acquisition

Significant work was done to achieve imaging by way of a spectrometer/PIXIS reading at each point in the intensity graph. Though this does yield extremely dense spectral information, the additional dwell time introduced by this kind of acquisition make it less than desirable. As an example, an exposure time of 2 seconds per spectra and a move / settle time of 100ms per point would yield a 100 x 100 pixel intensity graph requiring an average scan acquisition time of about 6 hours. This can be compared to the photodiode acquisition which often takes seconds per scan. In addition, recent work¹⁷ has shown an ability to achieve multi-modal imaging via spectral focusing onto a photodiode without the use of a spectrometer.

There are some tidbits of this project that may prove useful, however. For those measurements that are extremely broadband, this acquisition can resolve such pulses more accurately than a photodiode. In addition, the spectrometer/PIXIS can index multiple Raman shifts simultaneously, whilst the chirped multi-modal acquisition images multiple shifts in extremely rapid succession. This could be applied to in-situ experiments, allowing for the observation of a multitude of chemical changes occurring simultaneously. Also useful is a spectral analysis function developed, detailed here.

Acquisition via spectrometer follows much of the same preparation as the photodiode acquisition, creating all channels first and developing an XY raster waveform. It deviates in its scanning, which is software timed due to the camera's long exposure times. Most interesting is how it handles the data output of the PIXIS camera at each point, which is by nature a full spectrum. Translated into pixels and intensities, this enters into LabVIEW as a 1D array of

len(pixels), with a value at each pixel of the intensity. Here we demonstrate two options for handling such an array, which may prove useful for more traditional Raman and general spectroscopic experiments.

A user can first manually select the peak to be examined by entering in its pixel # and peak width (also in pixels) as obtained by the Lightfield software. The program then selects and sums the pixels in our array that correspond to said peak and pixels within its width, summing the values. This essentially translates to a Riemann sum. More excitingly, a user can have the program do such work for them, automatically finding the highest peak present in a spectra and integrating asymmetrically along the rising and falling edges of the peak.

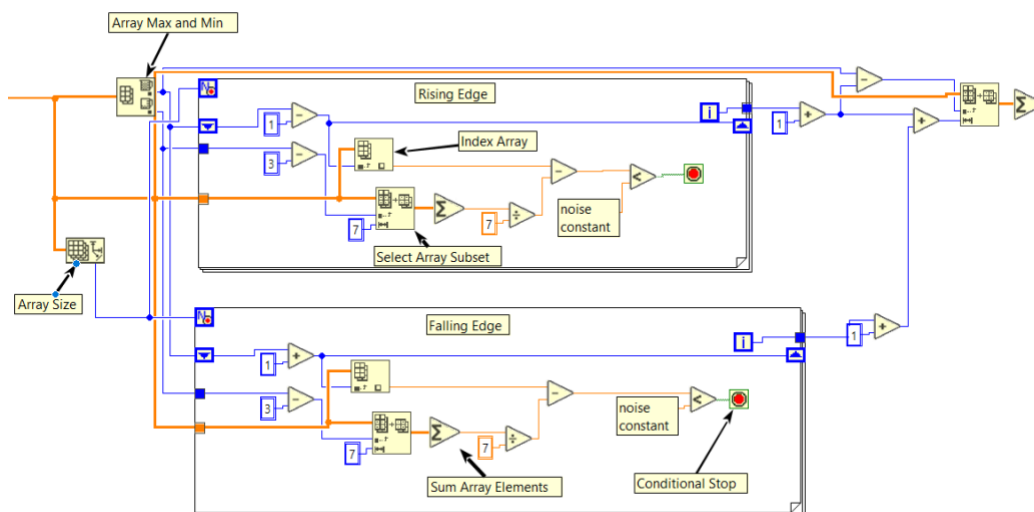


Figure 17: Block diagram of spectral analysis.vi

The structure begins with two for loops that have a maximum length of the entire array. It utilizes Array Max and Min to select the indices of our array that correspond to the peak (max) and baseline (min). For each side of the peak, it then iterate downwards until we hit the

baseline, then record how many pixels the peak encompasses. Examining the Rising Edge case, it utilizes a for loop to iterate one pixel to the left repeatedly until the intensity is roughly equivalent to the baseline as determined by a user-inputted noise constant. The value of the baseline is determined by taking the average value of a number of elements surrounding the spectral minimum. The same process is then repeated for the falling edge. Once the two widths have been acquired, the program sums over the whole peak, recording that as the intensity at that galvanometric point.

Conclusion & Future Work

With less than a year in development, the project is still very much in progress. We have successfully built, integrated, aligned, and developed software for: the Thorlabs—GVSK2 Galvanometric mirrors, the PIXIS Spectrometer, the Nikon Ti-2 and a 4f relay. Upon further verification of our pulse profile and implementation of an Electro-Optic Modulator, Lock-In Amplifier and Function Generator, we will have achieved stimulated Raman spectroscopy with a minimum pixel dwell time of 4 microseconds and a pulse timing of 60 fs, all controlled by an in-house designed software.

A number of steps remain. For an accurate measurement of pump on/off to achieve a pump loss scheme, an EOM must be implemented alongside a function generator to provide a steady frequency. With a relatively low laser repetition rate of 5 kHz, proper care must be taken to ensure synchronization of data acquisition, galvanometric motion, and pulse arrivals. A proposed TTL synchronization scheme can be seen in figure 18. The function generator will

output a 2.5 kHz square wave reference to both the EOM and lock-in amplifier, modulating our 5kHz laser on a pulse on/pulse off basis. The lock-in amplifier operates via a user inputted time constant, which is the amount of time the lock-in averages a DC signal over. By setting this time constant to 1/5 of the pixel dwell time, any weight from previous, non-pixels values is diminished, ensuring a steady and accurate signal. This signal input from the lock-in amplifier via analog input is acquired at the end of the pixel dwell time (falling edge).

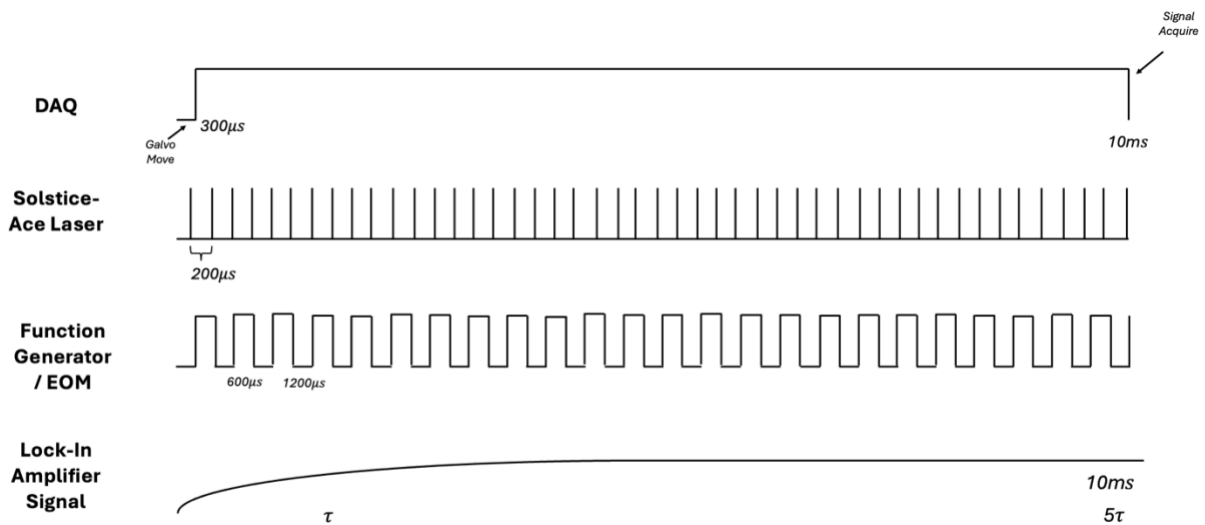


Figure 18: TTL synchronization scheme w/ example Lock-in output

References

- (1) Maier, O.; Oberle, V.; Hoekstra, D. Fluorescent Lipid Probes: Some Properties and Applications (a Review). *Chem. Phys. Lipids* **2002**, *116* (1–2), 3–18.
[https://doi.org/10.1016/S0009-3084\(02\)00017-8](https://doi.org/10.1016/S0009-3084(02)00017-8).
- (2) Raman, C. V.; Krishnan, K. S. A New Type of Secondary Radiation [11]. *Nature* **1928**, *121* (3048), 501–502. <https://doi.org/10.1038/121501c0>.
- (3) Jones, R. R.; Hooper, D. C.; Zhang, L.; Wolverson, D.; Valev, V. K. Raman Techniques: Fundamentals and Frontiers. *Nanoscale Res. Lett.* **2019**, *14* (1), 231.
<https://doi.org/10.1186/s11671-019-3039-2>.
- (4) Smith, E.; Dent, G. Modern Raman Spectroscopy-A Practical Approach.
- (5) PENNEY, C. M.; GOLDMAN, L. M.; LAPP, M. Raman Scattering Cross Sections. *Nature Physical Science* **1972**, *235*:58 **1972**, *235* (58), 110–112.
<https://doi.org/10.1038/physci235110b0>.
- (6) Koike, K.; Koike, K.; Smith, N. I.; Fujita, K.; Fujita, K.; Fujita, K. Spectral Focusing in Picosecond Pulsed Stimulated Raman Scattering Microscopy. *Biomedical Optics Express*, *Vol. 13, Issue 2, pp. 995-1004* **2022**, *13* (2), 995–1004.
<https://doi.org/10.1364/boe.445640>.
- (7) Hu, F.; Shi, L.; Min, W. Biological Imaging of Chemical Bonds by Stimulated Raman Scattering Microscopy. *Nature Methods* **2019**, *16*:9 **2019**, *16* (9), 830–842.
<https://doi.org/10.1038/s41592-019-0538-0>.
- (8) Heuke, S.; Lombardini, A.; Lombardini, A.; Büttner, E.; Rigneault, H.; Rigneault, H. Simultaneous Stimulated Raman Gain and Loss Detection (SRGAL). *Optics Express*, *Vol. 28, Issue 20, pp. 29619-29630* **2020**, *28* (20), 29619–29630.
<https://doi.org/10.1364/OE.400298>.
- (9) Snow, J. B.; Qian, S.-X.; Chang, R. K. Stimulated Raman Scattering from Individual Water and Ethanol Droplets at Morphology-Dependent Resonances. *Optics Letters*, *Vol. 10, Issue 1, pp. 37-39* **1985**, *10* (1), 37–39. <https://doi.org/10.1364/ol.10.000037>.
- (10) Folick, A.; Min, W.; Wang, M. C. Label-Free Imaging of Lipid Dynamics Using Coherent Anti-Stokes Raman Scattering (CARS) and Stimulated Raman Scattering (SRS) Microscopy. *Curr. Opin. Genet. Dev.* **2011**, *21* (5), 585–590.
<https://doi.org/10.1016/J.GDE.2011.09.003>.

- (11) Brzozowski, K.; Matuszyk, E.; Pieczara, A.; Firlej, J.; Nowakowska, A. M.; Baranska, M. Stimulated Raman Scattering Microscopy in Chemistry and Life Science – Development, Innovation, Perspectives. *Biotechnol. Adv.* **2022**, *60*, 108003. <https://doi.org/10.1016/J.BIOTECHADV.2022.108003>.
- (12) Qian, N.; Gao, X.; Lang, X.; Deng, H.; Bratu, T. M.; Chen, Q.; Stapleton, P.; Yan, B.; Min, W. Rapid Single-Particle Chemical Imaging of Nanoplastics by SRS Microscopy. *Proceedings of the National Academy of Sciences* **2024**, *121* (3), e2300582121. <https://doi.org/10.1073/PNAS.2300582121>.
- (13) Patton, B. Using an Autocollimator to Align 4f Systems. **2022**. <https://doi.org/10.1242/FOCALPLANE.11048>.
- (14) Rezai, M.; Salehi, J. A. Fundamentals of Quantum Fourier Optics. *IEEE Transactions on Quantum Engineering* **2023**, *4*. <https://doi.org/10.1109/TQE.2022.3224799>.
- (15) *The story of NI and its influence on instrument control - EDN*. <https://www.edn.com/the-story-of-ni-and-its-influence-on-instrument-control/> (accessed 2026-04-03).
- (16) Ferrand, P. GPScan.VI: A General-Purpose LabVIEW Program for Scanning Imaging or Any Application Requiring Synchronous Analog Voltage Generation and Data Acquisition. *Comput. Phys. Commun.* **2015**, *192*, 342. <https://doi.org/10.1016/J.CPC.2015.03.010>.
- (17) Liao, C.-S.; Huang, K.-C.; Hong, W.; Chen, A. J.; Karanja, C.; Wang, P.; Eakins, G.; Cheng, J.-X. Stimulated Raman Spectroscopic Imaging by Microsecond Delay-Line Tuning. *Optica*, Vol. 3, Issue 12, pp. 1377-1380 **2016**, *3* (12), 1377–1380. <https://doi.org/10.1364/optica.3.001377>.

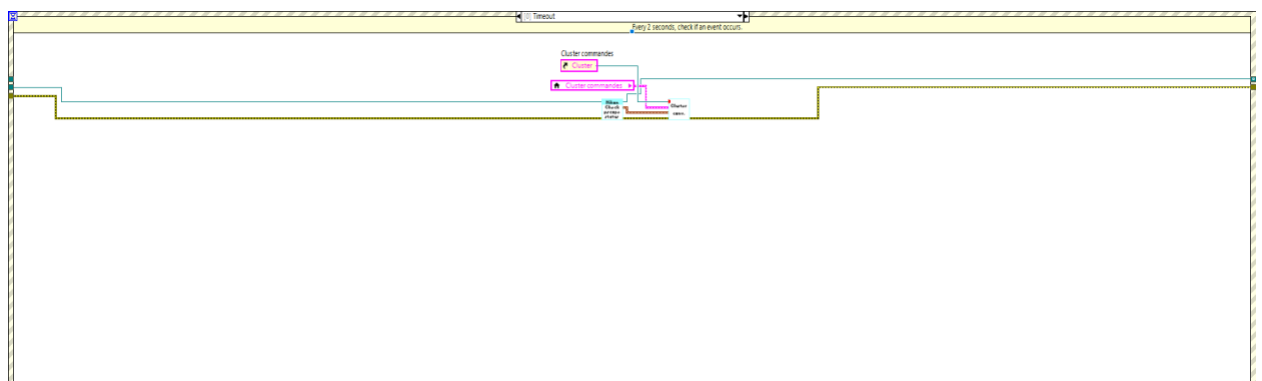
Appendix

Data Acquisition Card Casing



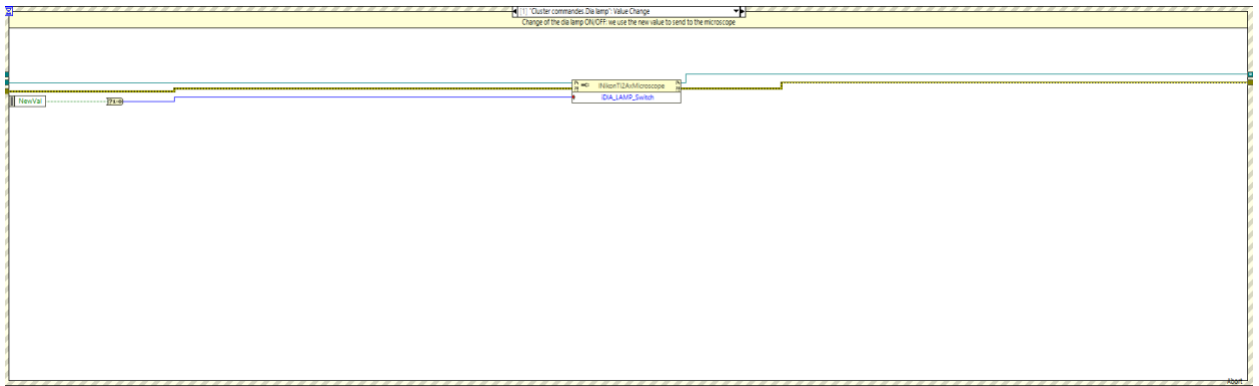
Event Cases

0. Timeout (default)



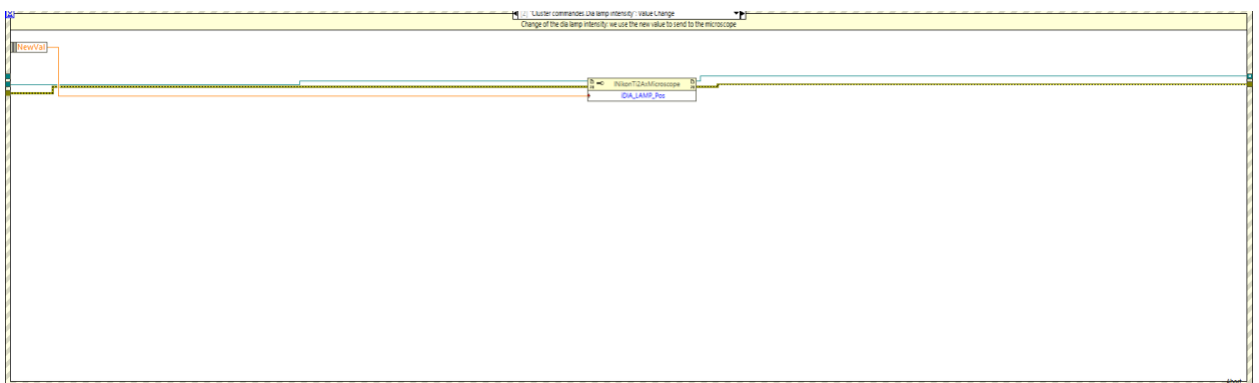
Checks/updates microscope values

1. "Cluster commands. Dia lamp": Value Change



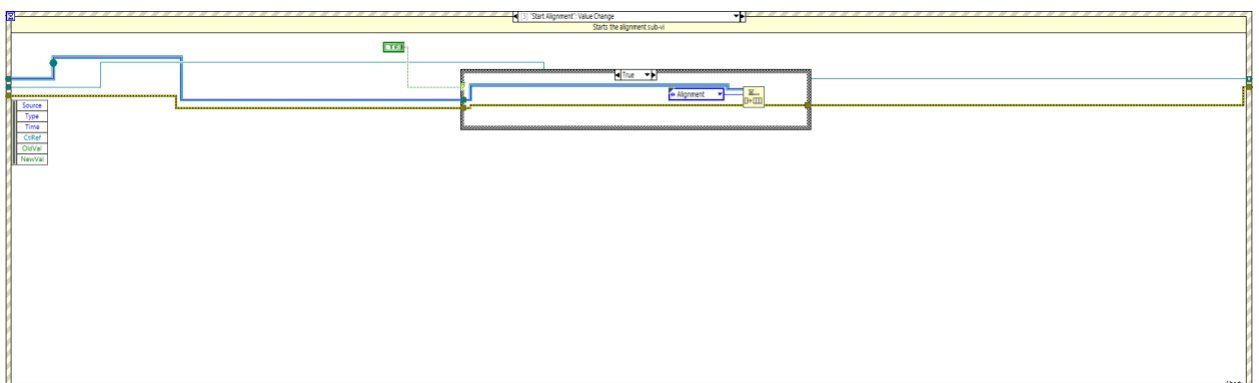
Switches Lamp On/Off

2. “Cluster commands. Dia lamp intensity”: Value Change



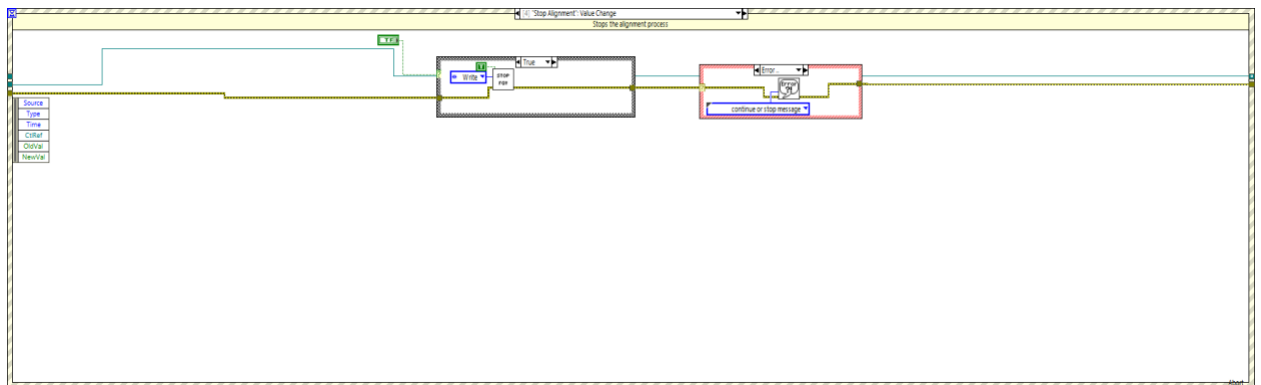
Changes intensity of microscope lamp

3. “Start Alignment”: Value Change



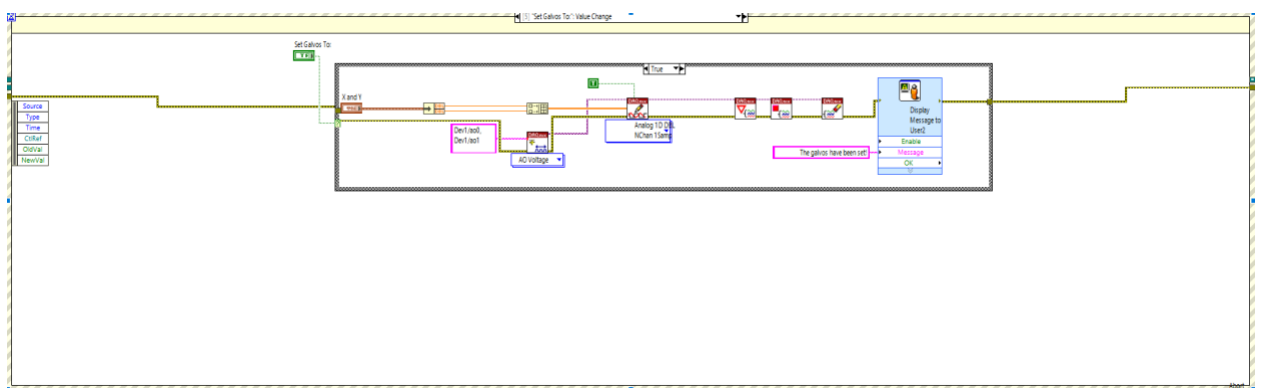
Enqueues Alignment command to consumer loop

4. “Stop Alignment”: Value Change



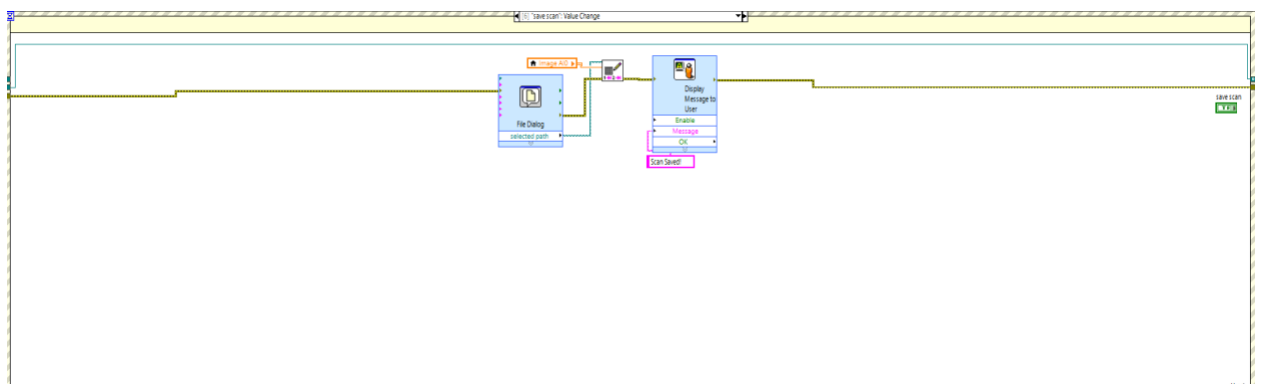
Writes True to the STOP FGV, which halts Alignment.vi

5. "Set Galvos To": Value Change



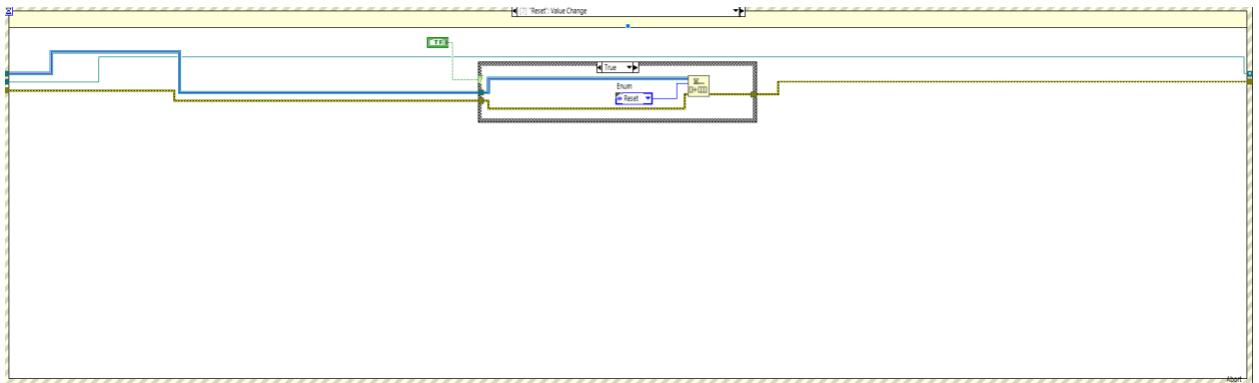
Moves Galvos to specific voltage/angle

6. "save scan": Value Change



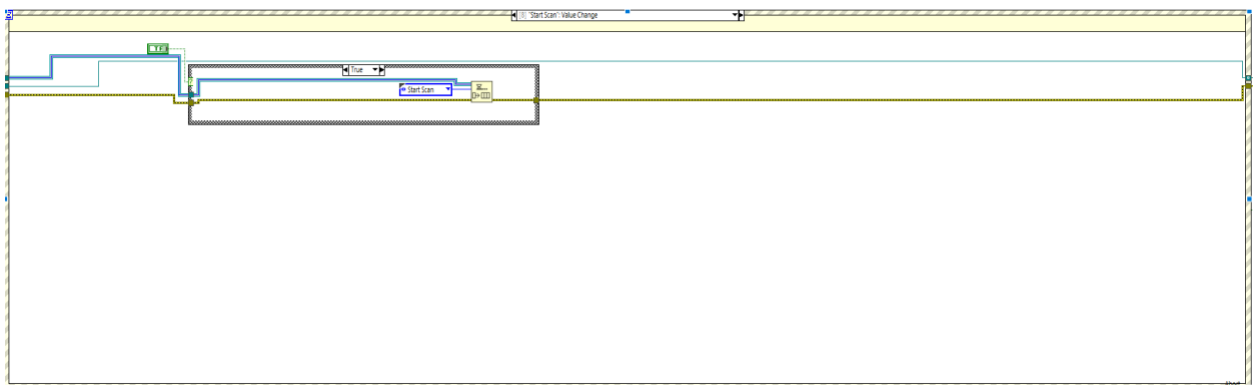
Saves Scan to User-Inputted File Location

7. "Reset": Value Change



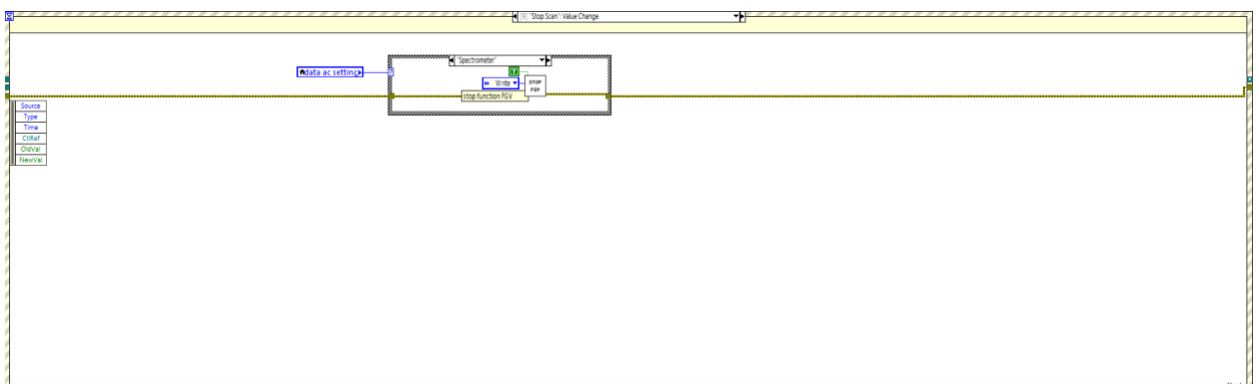
Enqueues Reset command to consumer loop

8. "Start Scan": Value Change



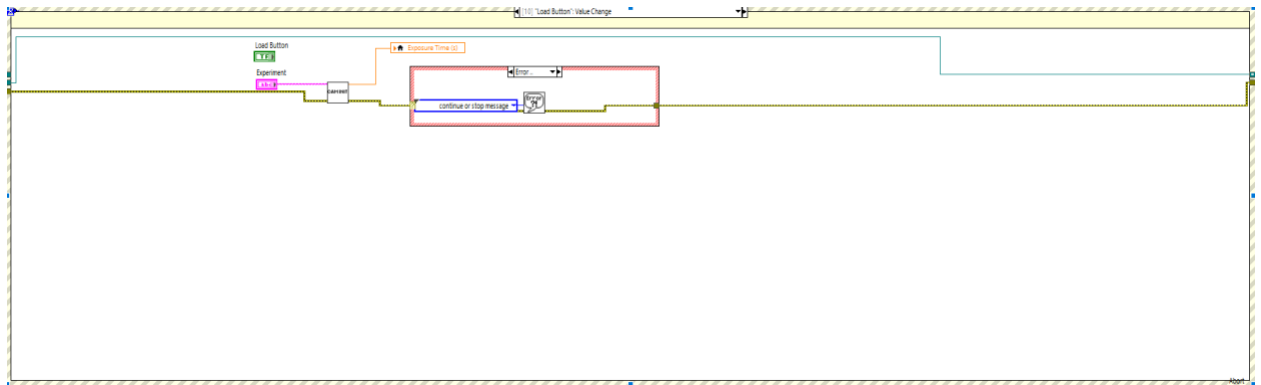
Enqueues Start command to consumer loop

9. "Stop Scan": Value Change



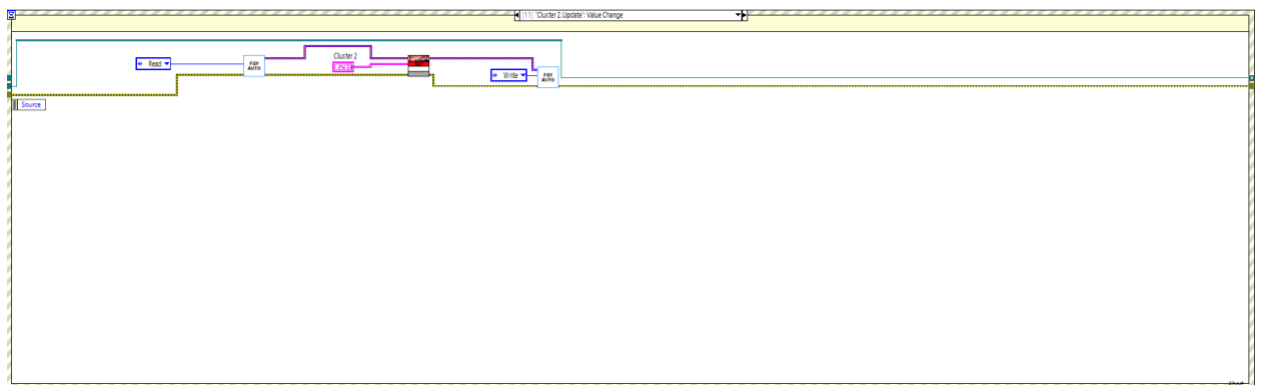
Writes true to STOP FGV if on spectrometer, else idle

10. "Load Button": Value Change



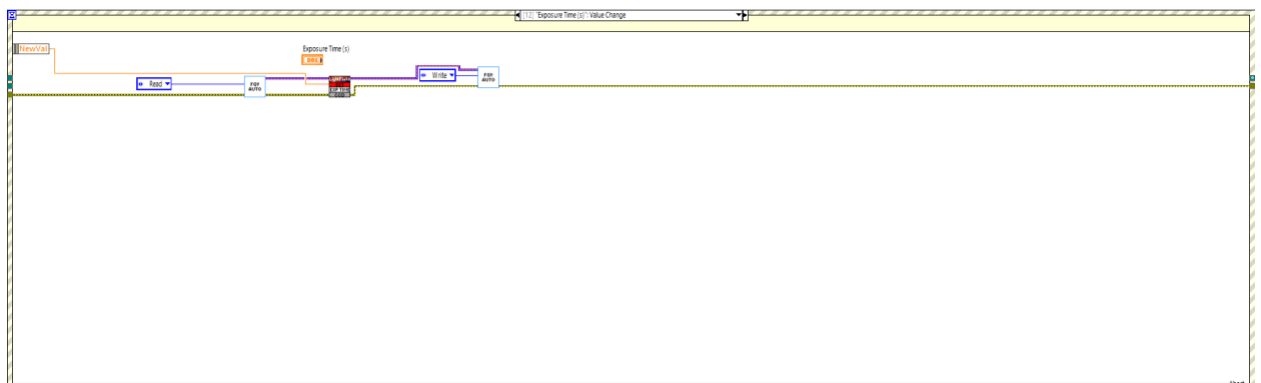
Loads experiment using Cam Init.vi

11. "Region Of Interest.Update": Value Change



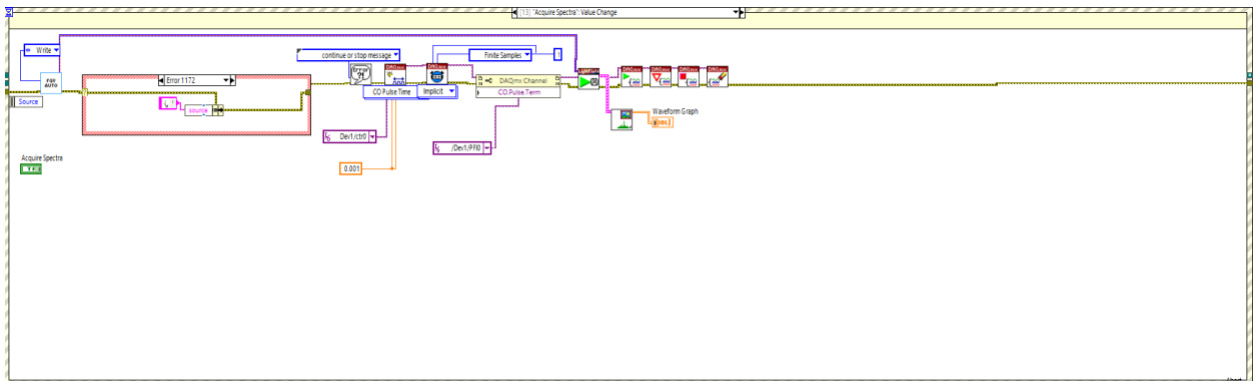
Loads custom ROI using Set_ROI.vi

12. "Exposure Time (s)": Value Change



Sets exposure time

13. "Acquire Spectra": Value Change



Acquires spectra via spectrometer

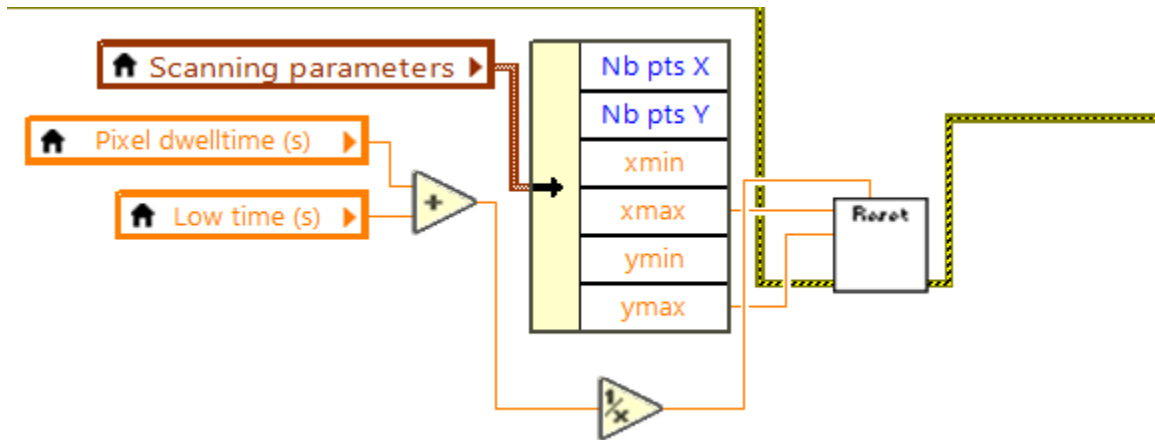
14. "Save Spectra": Value Change



Saves spectra to user-inputted file path

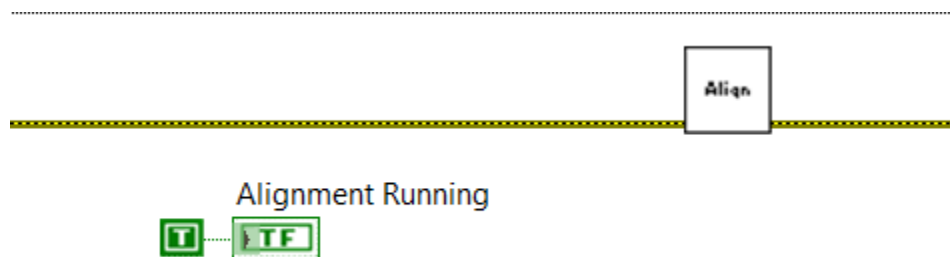
Consumer Loop

1. Idle (default)
2. "Reset"



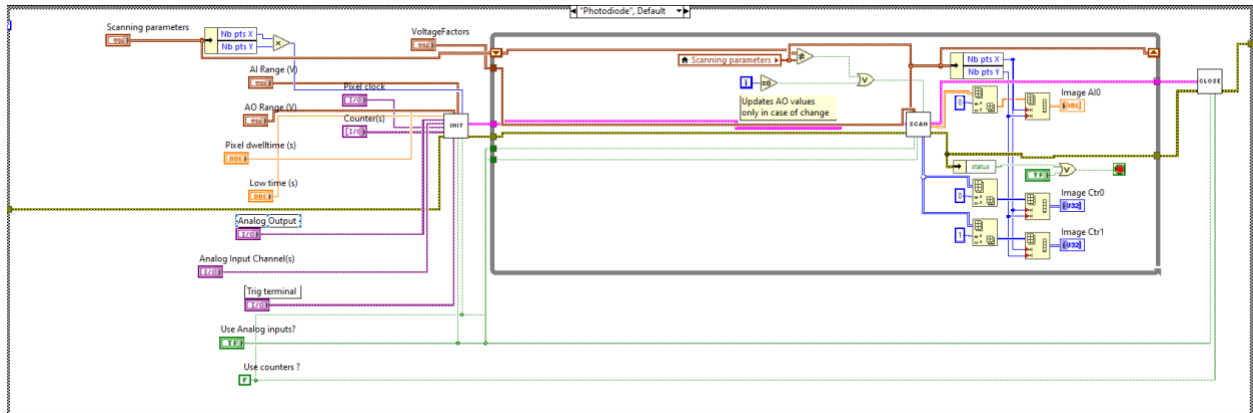
Runs Reset.vi

3. Alignment

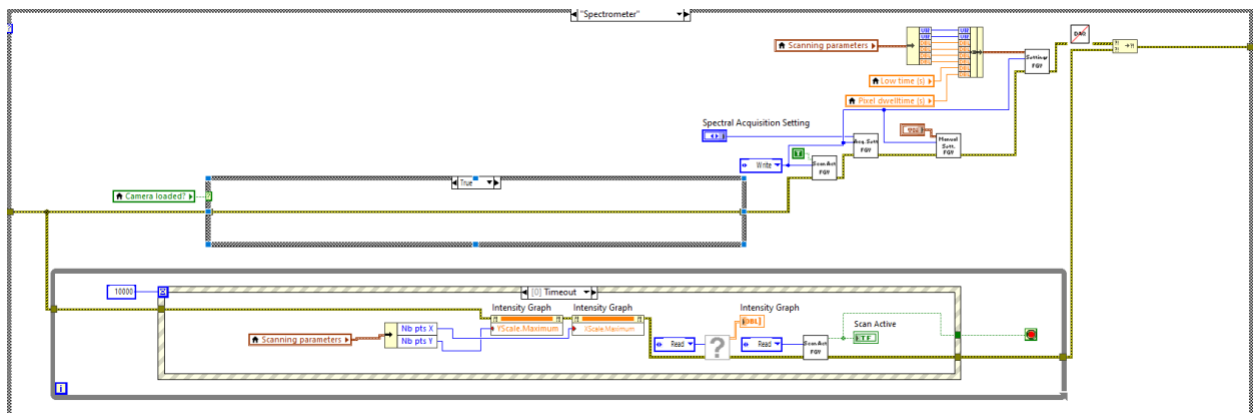


Sets Boolean, runs Alignment.vi

4. Start Scan

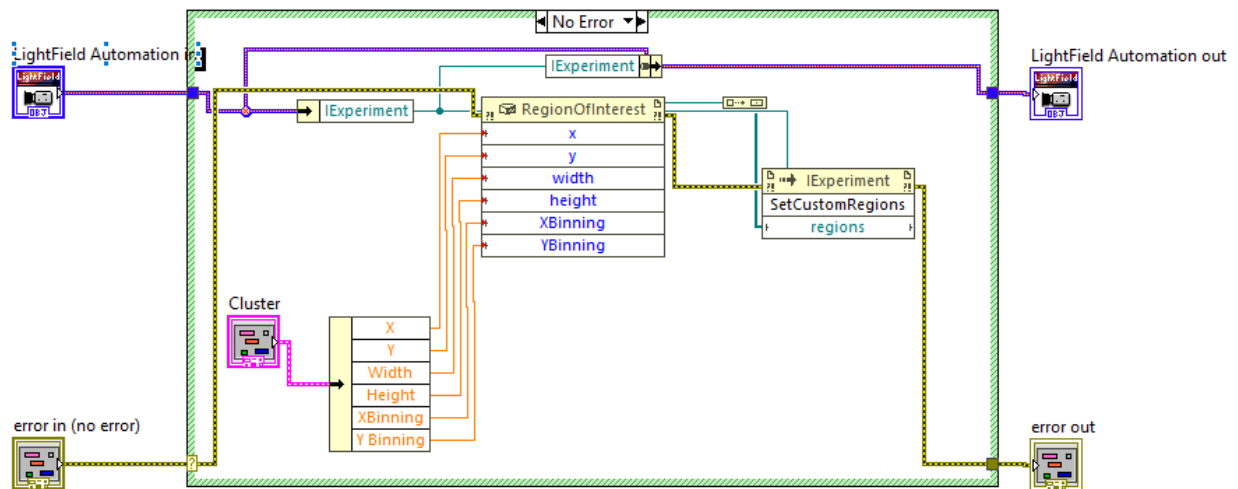


Runs photodiode acquisition

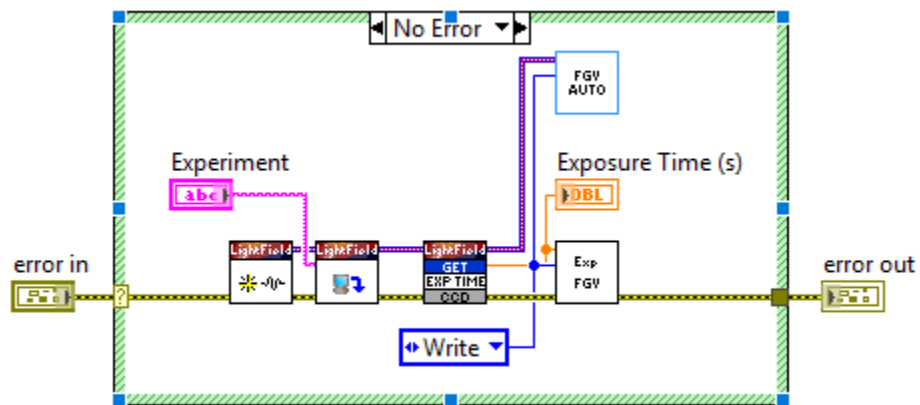


Runs spectrometer acquisition

Spectrometer Sub-Vis



Set_ROI.vi



Cam_Init.vi

Common Errors

1. The NI-DAQ has a tendency to disconnect itself when the computer has been in sleep mode for extended periods of time. While not displaying an obvious error, this causes galvo commands to be moot
 - a. The USB-C connector should either be disconnected and reconnected after startup.
2. The Lightfield software is extremely finicky, as it locks any hardware utilized in an experiment as soon as it is loaded. Upon trying to load another experiment, the user may encounter an error detailing this
 - a. If possible, experimental loads should be limited to once per Microscope.vi run. If another instance of Lightfield is open elsewhere, it should be terminated beforehand.
3. The Microscope integration requires a pre-existing “hardware tag” (reference) to control the microscope remotely.
 - a. The Nikon Ti-2 should be powered on before the LabVIEW is launched.
4. Unfortunately, the galvos experience some drift over time. This can be seen in a misalignment to the microscope objective.
 - a. Reset.vi or Alignment.vi can be used to return the galvos to their (0,0) position.

The mirror before the galvos can also be adjusted slightly, as well as the rotation of the galvanometric motors for each dimension.